The Dot Product

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Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Computing dot products and angles

Let $\mathbf{u} = \langle 2, 5 \rangle$ and $\mathbf{v} = \langle -3, 4 \rangle$. Find:

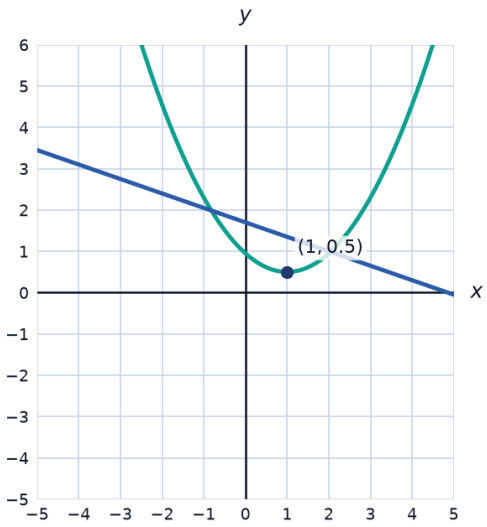
a $\mathbf{u} \cdot \mathbf{v}$

b $|\mathbf{u}|$ and $|\mathbf{v}|$

c the angle between $\mathbf{u}$ and $\mathbf{v}$, to one decimal place

2 [Foundational] • Target: Computing dot products and angles

Determine whether each pair of vectors is orthogonal:



a $\langle 4, -6 \rangle$ and $\langle 3, 2 \rangle$

b $\langle 1, 2 \rangle$ and $\langle 2, 1 \rangle$

3 [Foundational] • Target: Computing dot products and angles

Find the value of k so that $\langle k, 3 \rangle$ is orthogonal to $\langle 2, -4 \rangle$.

4 [Foundational] • Target: Projections and work

A force $\mathbf{F} = \langle 8, 5 \rangle$ N moves an object along the displacement $\mathbf{d} = \langle 10, 0 \rangle$ m. Find the work done.

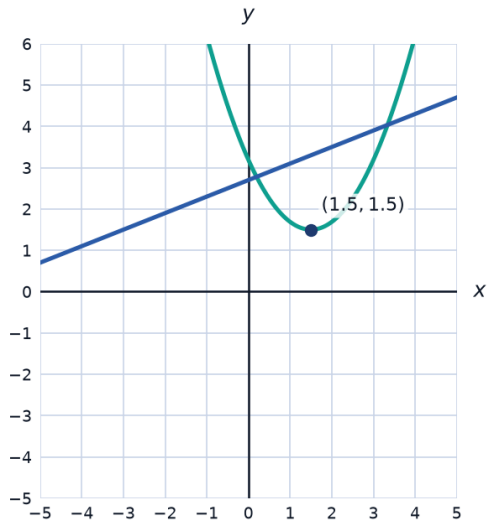
5 [Foundational] • Target: Projections and work

A wagon is pulled a distance of 20 m by a rope making an angle of $30°$ with the ground, with a force of magnitude 50 N. Find the work done, to one decimal place.

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6 [Foundational] • Target: Properties of the dot product

Let $\mathbf{u} = \langle 3, 1 \rangle$, $\mathbf{v} = \langle -2, 4 \rangle$, $\mathbf{w} = \langle 1, 1 \rangle$. Verify that $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$ (the distributive property).



7 [Foundational] • Target: Angle classification

Without finding the angle exactly, classify each pair of vectors as forming an acute, right, or obtuse angle, using only the sign of the dot product:

a $\langle 3, 5 \rangle$ and $\langle 2, -1 \rangle$

b $\langle -1, 4 \rangle$ and $\langle 8, 2 \rangle$

c $\langle 2, 2 \rangle$ and $\langle -3, -3 \rangle$

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Scalar projection and further applications

Find the scalar projection of $\mathbf{u} = \langle 5, -2 \rangle$ onto $\mathbf{v} = \langle 3, 4 \rangle$ (the signed length $\text{comp}_{\mathbf{v}}\mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|}$), to two decimal places.

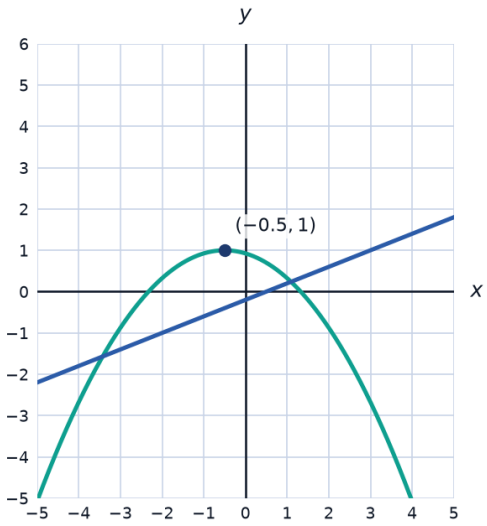
9 [Intermediate] • Target: Scalar projection and further applications

A sled is pulled by a rope at $40°$ above horizontal with a force of 120 N, moving the sled 15 m horizontally. Find the work done by the horizontal component of the force.

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10 [Intermediate] • Target: Decomposing a vector into parallel and perpendicular parts

Decompose $\mathbf{u} = \langle 7, 4 \rangle$ into a component parallel to $\mathbf{v} = \langle 1, 0 \rangle$ and a component perpendicular to $\mathbf{v}$.



11 [Intermediate] • Target: Deriving the dot product angle formula

State the formula relating the dot product to the angle between two vectors, and explain why $\cos\theta = 0$ corresponds to orthogonal vectors.

12 [Intermediate] • Target: The Cauchy-Schwarz inequality (preview)

Explain why $|\mathbf{u} \cdot \mathbf{v}| \leq |\mathbf{u}||\mathbf{v}|$ must always hold, using the formula $\mathbf{u} \cdot \mathbf{v} = |\mathbf{u}||\mathbf{v}|\cos\theta$.

13 [Intermediate] • Target: Dot products with variable components

Find all values of k such that $\langle k, 3 \rangle$ and $\langle k, -3 \rangle$ are orthogonal.

14 [Intermediate] • Target: Dot products with variable components

Find all values of k such that the angle between $\langle 1, k \rangle$ and $\langle 2, 1 \rangle$ is $45°$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

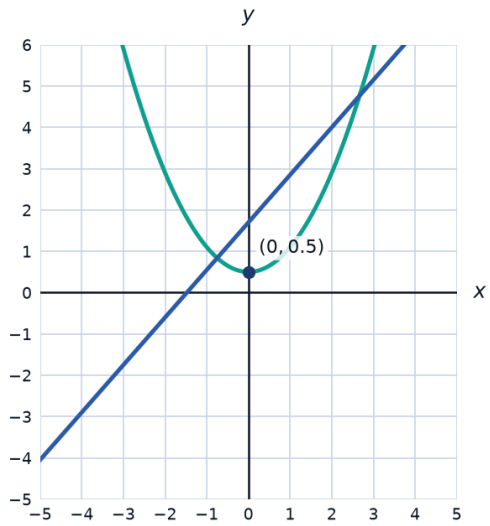
15 [Challenging] • Target: A capstone dot product application • Cross-Lesson: 1.02 Domain and Range

A weather vane experiences wind force $\mathbf{F} = \langle 15, -8 \rangle$ N. The vane can only rotate about a pivot, so only the component of force perpendicular to the vane's current direction $\mathbf{d} = \langle 3, 4 \rangle$ causes it to turn. Find the magnitude of the perpendicular (turning) component of the force, using the fact that the perpendicular component's magnitude equals $\frac{|\mathbf{F} \times \mathbf{d}|}{|\mathbf{d}|}$ in 2D (using the scalar cross product $F_x d_y - F_y d_x$), and separately confirm the parallel component using the dot product.

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16 [Challenging] • Target: A capstone dot product application • Cross-Lesson: 1.03 Average Rate of Change

Two salespeople's monthly sales across three product categories are given by vectors $\mathbf{s}_1 = \langle 40, 25, 10 \rangle$ and $\mathbf{s}_2 = \langle 15, 30, 45 \rangle$ (units sold). Find $\mathbf{s}_1 \cdot \mathbf{s}_2$ and explain what this dot product does NOT directly tell you about their relative performance (versus what the individual sums would show).



17 [Challenging] • Target: Deriving the dot product angle formula • Cross-Lesson: 2.01 Right-Triangle Trigonometry

Level 2 · MediumVector $\vec{u} = \langle 4, -3 \rangle$ and vector $\vec{v} = \langle 1, 7 \rangle$. Calculate the magnitude of each vector, determine the dot product $\vec{u} \cdot \vec{v}$, and find the exact angle θ between them in radians.

18 [Challenging] • Target: The Cauchy-Schwarz inequality (preview) • Cross-Lesson: 2.02 Radian Measure and Arc Length

Find the exact angle θ between $\mathbf{u} = \langle 3, 1 \rangle$ and $\mathbf{v} = \langle 1, 2 \rangle$, and compute the vector projection $\text{proj}_{\mathbf{v}}\mathbf{u}$.

19 [Challenging] • Target: Dot products with variable components • Cross-Lesson: 2.03 Graphs of Sine and Cosine Functions

[Sequences, Series, Induction & the Binomial Theorem · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

20 [Challenging] • Target: A capstone dot product application • Cross-Lesson: 2.04 Law of Sines and Law of Cosines

[Trigonometry & Triangle Laws · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Computing dot products and angles

Lines $\ell_1 \parallel \ell_2$ cut by a transversal form corresponding angles of $(4x - 3)^\circ$ and $(3x + 4)^\circ$. Find the measure of the angle $(4x - 3)^\circ$.

a. 26

b. 34

c. 25

d. 20

22 [Foundational] • Target: Projections and work

In ΔABC, the side lengths opposite to angles A, B, and C are a, b, and c, respectively. If $\angle A - \angle B = 90^\circ$ and $a + b = \sqrt{2}c$, what is the measure of $\angle C$?

a. 30°

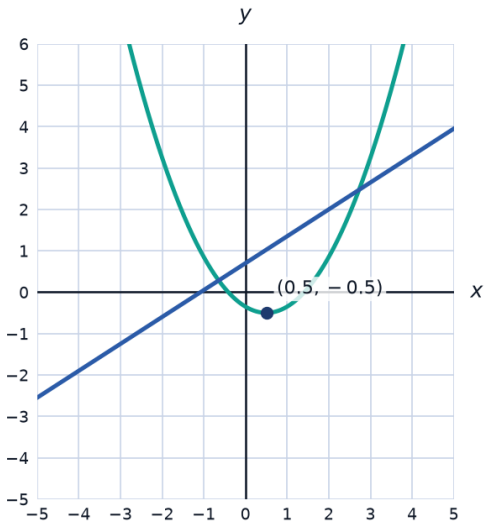
b. 45°

c. 60°

d. 75°

23 [Foundational] • Target: Properties of the dot product

In the diagram below, line AB is parallel to line CD. Transversal line EF intersects AB at point G and CD at point H. If the measure of angle AGH is 50°, what is the measure of angle GHD?



a. 50°

b. 130°

c. 40°

d. 90°

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24 [Foundational] • Target: Angle classification

In the figure, right triangle ABC has a right angle at vertex C, and altitude CD is drawn perpendicular to hypotenuse AB. If $AD = x$, $DB = x + 5$, and $CD = 6$, what is the length of hypotenuse AB?

a. 4

b. 12

c. 13

d. 23

25 [Foundational] • Target: Scalar projection and further applications

A particle of mass 5 kg moves under the action of a variable force $\mathbf{F}$ N. The force is given by $\mathbf{F} = (3x^2 - 2x)\mathbf{i}$ N, where x is the displacement in metres from a fixed origin O. The particle moves from $x = 2$ m to $x = 4$ m. Find the work done by the force $\mathbf{F}$ during this displacement.

a. 32 J

b. 44 J

c. 56 J

d. 68 J

26 [Foundational] • Target: Decomposing a vector into parallel and perpendicular parts

In the diagram below, lines AB and CD are parallel. Line EF is a transversal intersecting AB at G and CD at H. If the measure of angle AGH is $(2x + 10)^\circ$ and the measure of angle GHD is $(3x - 5)^\circ$, find the value of x.

a. 15

b. 25

c. 35

d. 45

27 [Foundational] • Target: Deriving the dot product angle formula

In the figure below, ΔABC is a right triangle with right angle at B, and altitude $\overline{BD}$ is drawn perpendicular to hypotenuse $\overline{AC}$. If $AD = 4$ and $DC = 9$, what is the length of $\overline{BD}$?

a. $\sqrt{13}$

b. 6

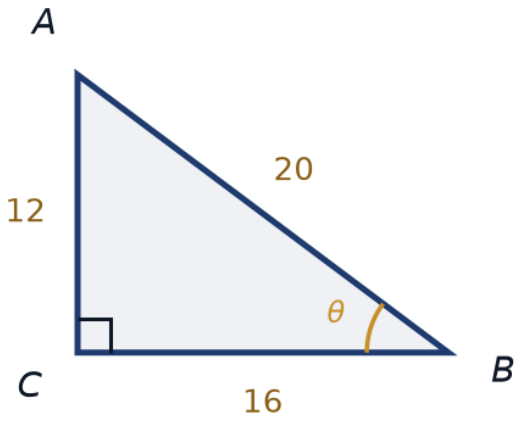
c. 6.5

d. 13

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28 [Foundational] • Target: The Cauchy-Schwarz inequality (preview)

Prove the product formula: $\cos A\cos B = \frac{1}{2}[\cos(A-B)+\cos(A+B)]$.



a. Only if $A = B$

b. Use integration

c. Not true

d. From $\cos(A-B) = \cos A\cos B + \sin A\sin B$ and $\cos(A+B) = \cos A\cos B - \sin A\sin B$. Add and halve. ✓

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Dot products with variable components

Prove the sum-to-product formula: $\sin A + \sin B = 2\sin\frac{A+B}{2}\cos\frac{A-B}{2}$.

a. Not true

b. Only for $A = B$

c. Set $A = M+N, B = M-N$; then $\sin(M+N)+\sin(M-N) = 2\sin M\cos N$. Substitute $M = (A+B)/2, N = (A-B)/2$. ✓

d. Use calculus

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30 [Intermediate] • Target: A capstone dot product application

State the sum-to-product formula for $\cos A - \cos B$.

a. $2\cos\frac{A+B}{2}\cos\frac{A-B}{2}$

b. $2\sin\frac{A+B}{2}\cos\frac{A-B}{2}$

c. $2\cos\frac{A+B}{2}\sin\frac{A-B}{2}$

d. $-2\sin\frac{A+B}{2}\sin\frac{A-B}{2}$

31 [Intermediate] • Target: Computing dot products and angles

Using the product-to-sum formula, simplify $\sin A\cos B$.

a. $\cos(A-B)$

b. $\frac{1}{2}[\sin(A+B)+\sin(A-B)]$

c. $\sin(A+B)$

d. $\frac{1}{2}[\cos(A-B)-\cos(A+B)]$

32 [Intermediate] • Target: Projections and work

Angles S and T are supplementary. If $\angle S = 34^\circ$, find $\angle T$.

a. 139

b. 143

c. 145

d. 146

33 [Intermediate] • Target: Properties of the dot product

Angles M and N are complementary. If $\angle M = 69^\circ$, find $\angle N$.

a. 21

b. 25

c. 16

d. 18

34 [Intermediate] • Target: Angle classification

Compute the dot product $\mathbf{u}\cdot\mathbf{v}$ for $\mathbf{u}=\langle 2,3\rangle$ and $\mathbf{v}=\langle 2,5\rangle$.

a. 19

b. 22

c. 16

d. 14

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35 [Intermediate] • Target: Scalar projection and further applications

Lines $\ell_1 \parallel \ell_2$ are cut by a transversal. Two co-interior (same-side interior) angles measure $(4x + 4)^\circ$ and 120°. Solve for x.

a. 21

b. 8

c. 20

d. 14

36 [Intermediate] • Target: Decomposing a vector into parallel and perpendicular parts

Two lines intersect, forming vertical angles. If one angle measures 114°, what is the measure of its vertical angle?

a. 116

b. 120

c. 119

d. 114

37 [Intermediate] • Target: Deriving the dot product angle formula

Given matrix $A = (2, -1; 3, 3)$, what is the scalar product $3A$?

a. $(5, 2; 6, 6)$

b. $(6, -1; 3, 9)$

c. $(6, -3; 9, 9)$

d. $(6, 3; 9, 10)$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: The Cauchy-Schwarz inequality (preview) • Cross-Lesson: 2.02 Radian Measure and Arc Length

Lines $\ell_1 \parallel \ell_2$ are cut by a transversal. One angle measures $(3x + 5)^\circ$ and its corresponding angle measures 80°. Solve for x.

a. 33

b. 20

c. 25

d. 19

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39 [Challenging] • Target: Dot products with variable components • Cross-Lesson: 2.03 Graphs of Sine and Cosine Functions

Evaluate $\int_0^1 xe^x\,dx$ using integration by parts.

a. -1

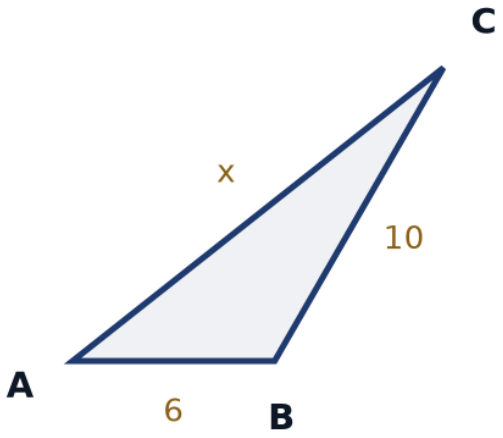
b. 1

c. $1 + e^2$

d. 2

40 [Challenging] • Target: A capstone dot product application • Cross-Lesson: 2.04 Law of Sines and Law of Cosines

In ΔABC shown below, side AB has length 6, side BC has length 10, and the area of the triangle is $15\sqrt{3}$. If $\angle B$ is an obtuse angle, what is the length of side AC, denoted by x?



a. 14

b. $2\sqrt{19}$

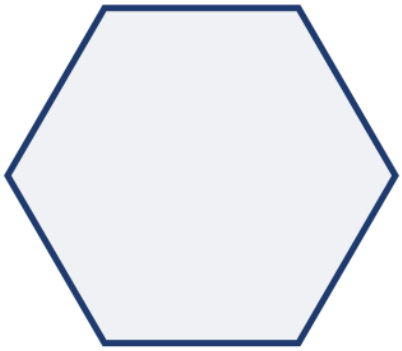
c. $2\sqrt{34}$

d. 16

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41 [Challenging] • Target: Computing dot products and angles • Cross-Lesson: 3.01 The Unit Circle and Exact Values

The sum of interior angles of a hexagon is:



a. 540°

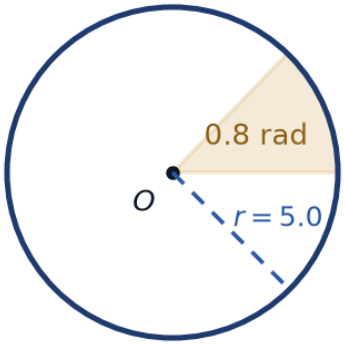
b. 720°

c. 900°

d. 360°

42 [Challenging] • Target: Projections and work • Cross-Lesson: 3.02 Fundamental Trigonometric Identities

A sector has radius 5 cm and angle 0.8 rad. Find the perimeter of the sector.



a. 14

b. 9

c. 4

d. 15

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43 [Challenging] • Target: Properties of the dot product • Cross-Lesson: 3.03 Sum and Difference Formulas

In the figure below, ΔABC is a right triangle with hypotenuse $\overline{AB}$, and altitude $\overline{CD}$ of length h is drawn to $\overline{AB}$. The altitude divides the hypotenuse into two segments of lengths $AD = 9$ and $DB = 16$. A circle with center O_1 is inscribed in ΔACD, and a circle with center O_2 is inscribed in ΔBCD. What is the distance between the centers O_1 and O_2?

a. 5

b. $5\sqrt{2}$

c. 7

d. $7\sqrt{2}$

44 [Challenging] • Target: Angle classification • Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations

In ΔABC, the side lengths opposite angles $A, B,$ and C are $a, b,$ and c, respectively. Given that $a = 7$, $b = 8$, and $\sin(B - A) = \frac{3}{5}\sin C,$ what is the length of side c?

a. 5

b. $3\sqrt{5}$

c. $\frac{5}{3}$

d. $2\sqrt{7}$

45 [Challenging] • Target: Scalar projection and further applications • Cross-Lesson: 4.01 Circles and Parabolas

A fair six-sided die is rolled twice. Let X be the random variable representing the sum of the outcomes of the two rolls. The probability distribution of X is given in the table below, where p is an unknown probability. $Sum\ (x), 2, 3, 4, 5, 6, 7; P(X = x), 136, p, 336, 436, 536, 636;$ What is the value of p?

a. $\frac{2}{36}$

b. $\frac{5}{36}$

c. $\frac{4}{36}$

d. $\frac{3}{36}$

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